



WHAT IS LEAN SIX SIGMA?



Lean focuses on removing wasteful activities in processes.

Six Sigma focuses on reducing process defects.

Leverages the best problem-solving methods to help organizations achieve their missions and satisfy customers.



Belt Levels of Lean Six Sigma



Who Benefits From Using Lean Six Sigma?

Anyone who has problems to solve including:

INDUSTRIES



Healthcare



Hi-Tech



Government



Retail



Finance



Hospitality

...Any Industry That Has Problems To Solve!

FUNCTIONS



Customer Service



Manufacturing



Accounting



Operations



Sales



HR & IT

...All Functions That Have Problems To Solve!

Lean Six Sigma Project Types



Quick Win

Implementation of a simple solution to a known issue

The problem is contained in one department, the root cause is known and the fix is painless

Also called "Just-Do-It" or Fast Track



Process Improvement

Incremental reduction of defects, cycle time or cost

The presenting issue has an unknown cause and solutions are not predetermined

Also called DMAIC, Lean or PDCA



Process Design

Creation of a brand-new, non-existent process

There is no existing process to analyze which requires benchmarking and collection of VOC

Also called DFSS or DMADV



Process Redesign

Overhaul of non-capable, existing process

The process exists, but incremental improvements will not be able to satisfy requirements

Also called Reengineering



Infrastructure Implementation

Establishment of key measurement systems

Monitoring of process capability and VOC are established to better focus improvement efforts

Also called Process Management

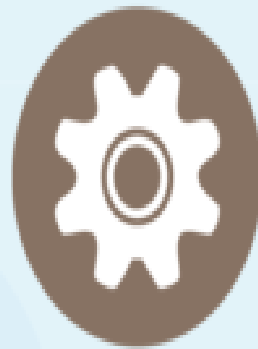
PDCA: Plan-Do-Check-Act

PDCA stands for Plan-Do-Check-Act.
It is a cyclical method for continuous improvement of processes.



PLAN

Create a process improvement plan.



DO

Execute a process improvement plan.



CHECK

Inspect feedback and adjust plan accordingly.



ACT

Integrate a process improvement plan into the system.

DMAIC

DMAIC is a five-step method for improving existing process problems with unknown causes.





Phase 1: Define

Define the problem. The Define Phase is the first phase of the Lean Six Sigma improvement process. In this phase the project team creates a Project Charter, a high-level map of the process and begins to understand the needs of the customers of the process. This is a critical phase in which the team outlines the project focus for themselves and the leadership of the organization.

Steps

Define the Problem by Developing a "Problem Statement"

Define the Goal by Developing a "Goal Statement"

Define the Process by Developing Process Maps

Define the Customer and Their Requirements

Inform Others of Project Progress

Tools

Project Charter

Tree Diagram

SIPOC

A3

Value Stream Map

Relationship Map

Swimlane Map

Stakeholder Analysis

Voice of the Customer (VOC) Translation Matrix

Project Charter

A Project Charter is a living document for an improvement team that outlines the presenting problem, the target and the boundaries of a process improvement effort.



Problem Statement

The problem captured in the form of a measurement



Business Case

The business reasons for doing the project



Goal Statement

The target of the process measurement



Timeline

When each project phase will be completed



Scope

What's in and what's out of the project



Team Members

The people who will participate in the project

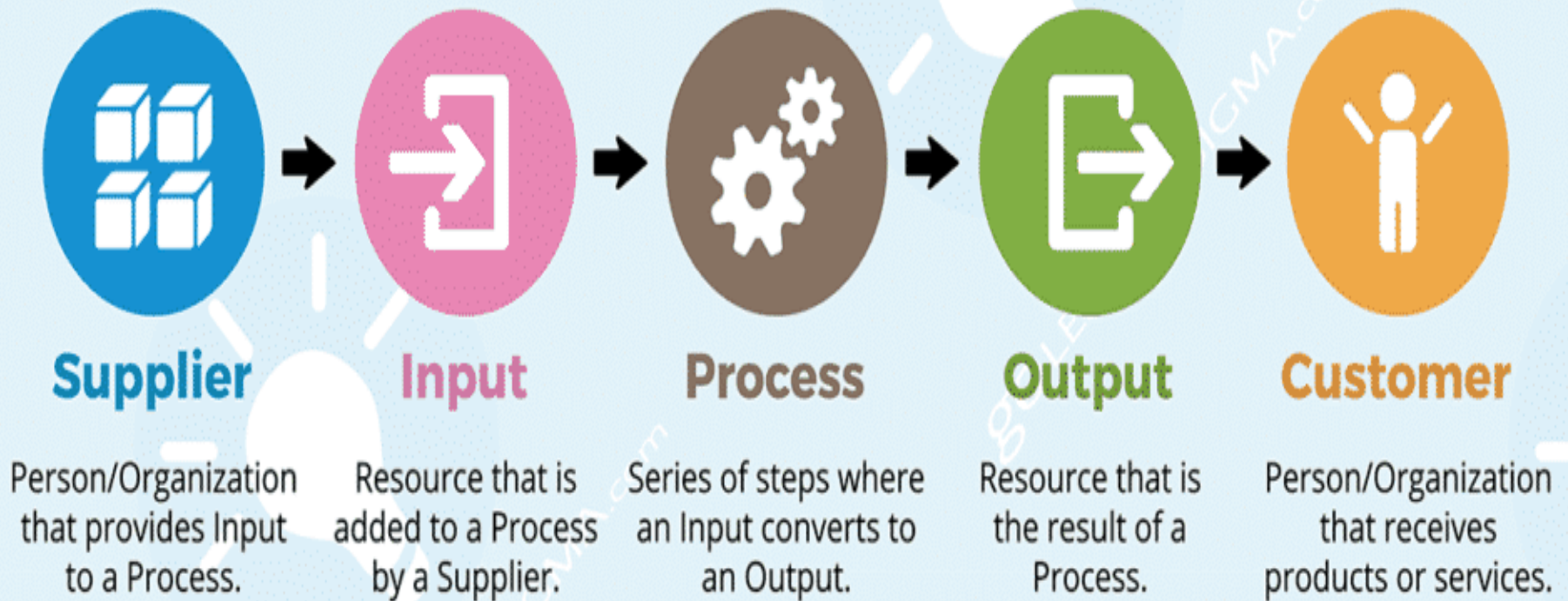
Project Selection

Project Selection is choosing a process improvement opportunity with the greatest impact and the most manageable effort while still aligning with organizational strategy.



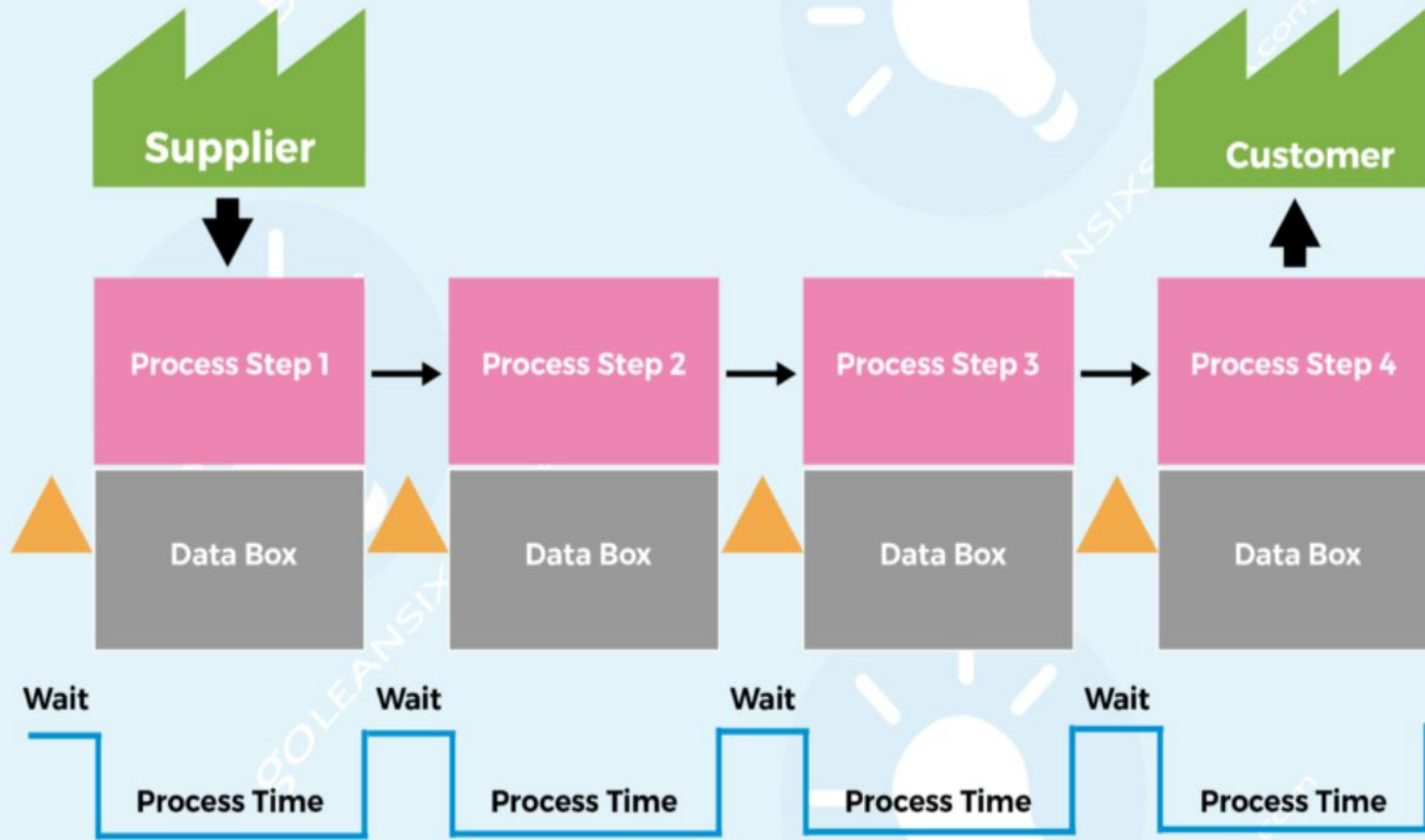
What Is a SIPOC?

A SIPOC is a high-level view of a process.
It stands for Suppliers, Inputs, Process, Outputs, and Customers:



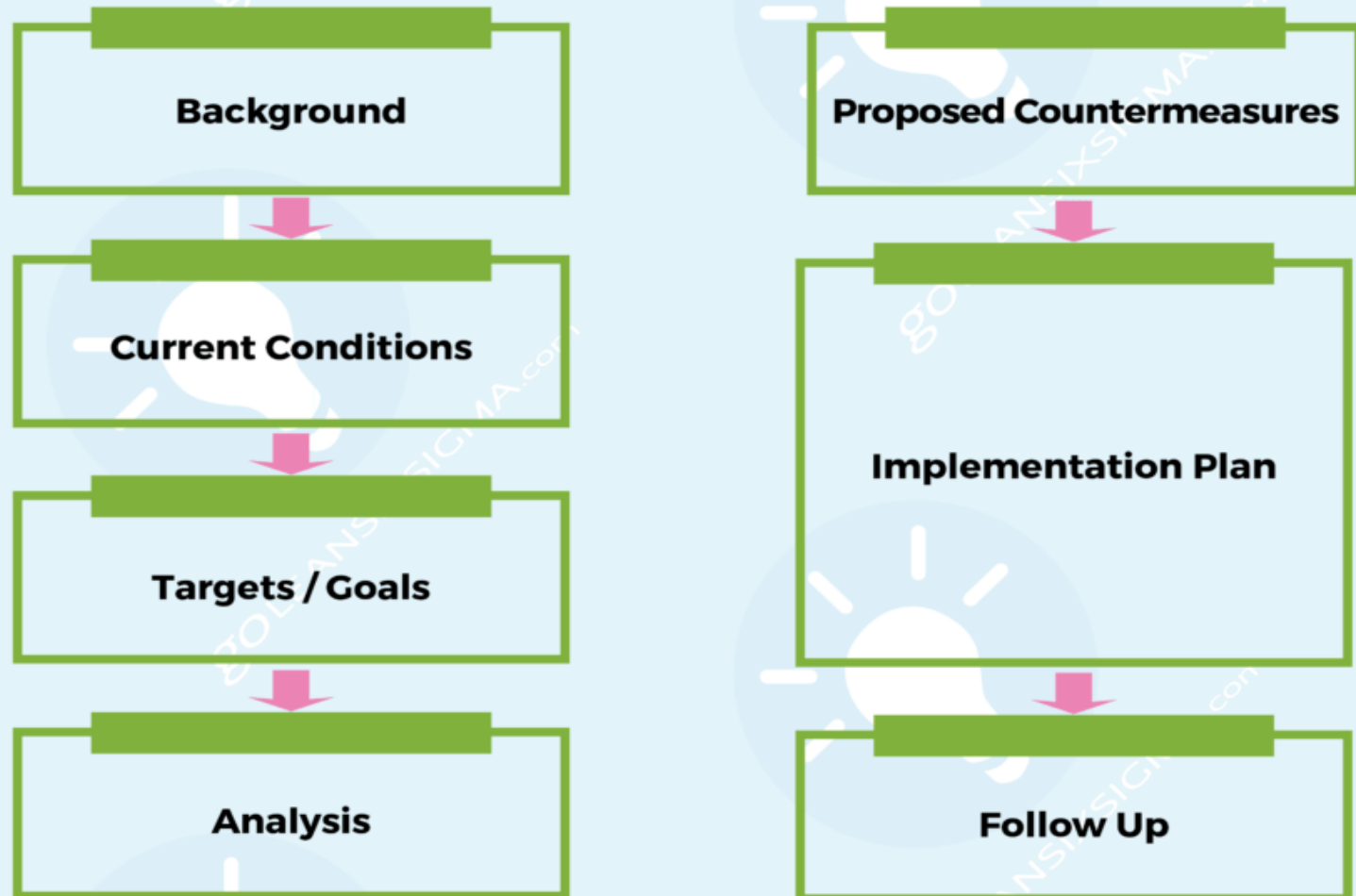
Value Stream Map

A Value Stream Map displays the high level process steps along with key process data.



A3

An A3 is a one-page report on the background and current state of a project.



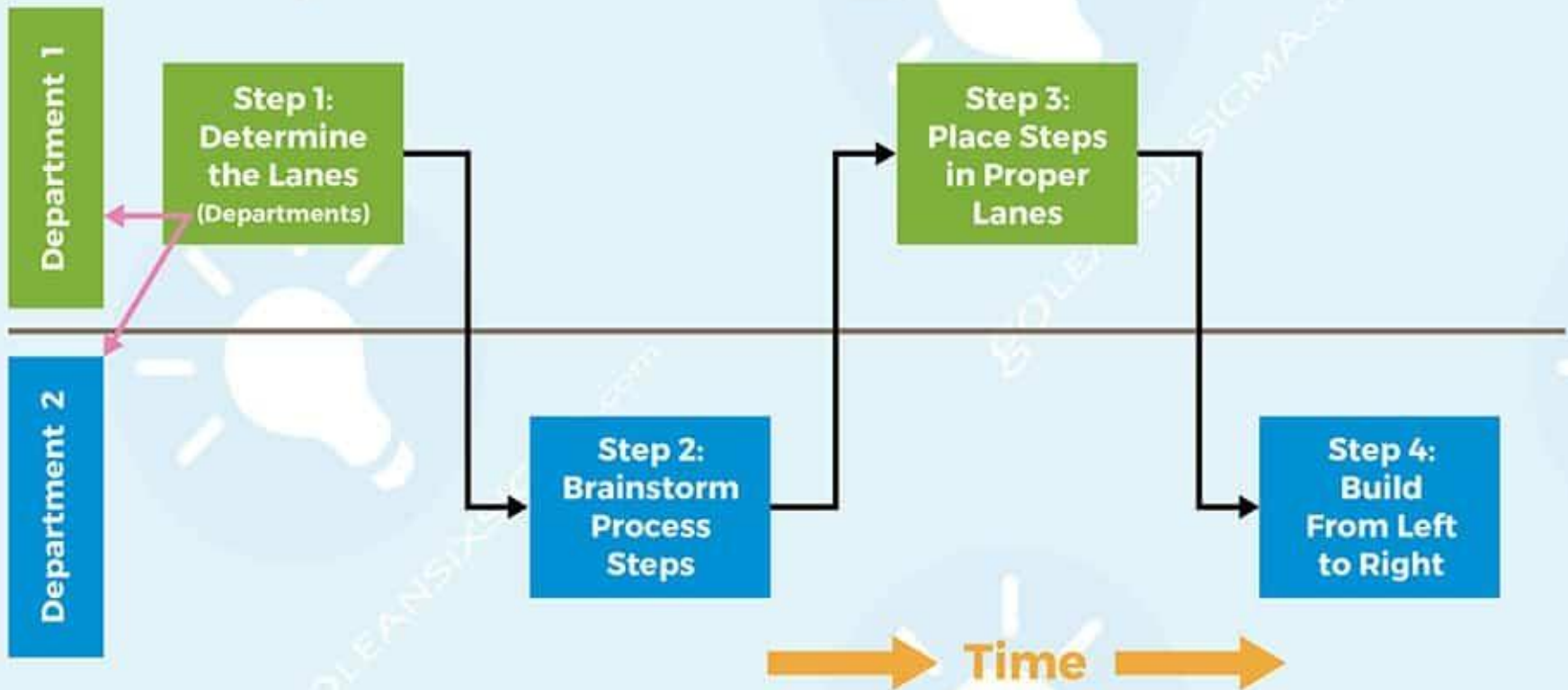
Stakeholder Analysis

Stakeholder Analysis is the process of understanding who has a vested interest in a change effort and working with them to ensure success.

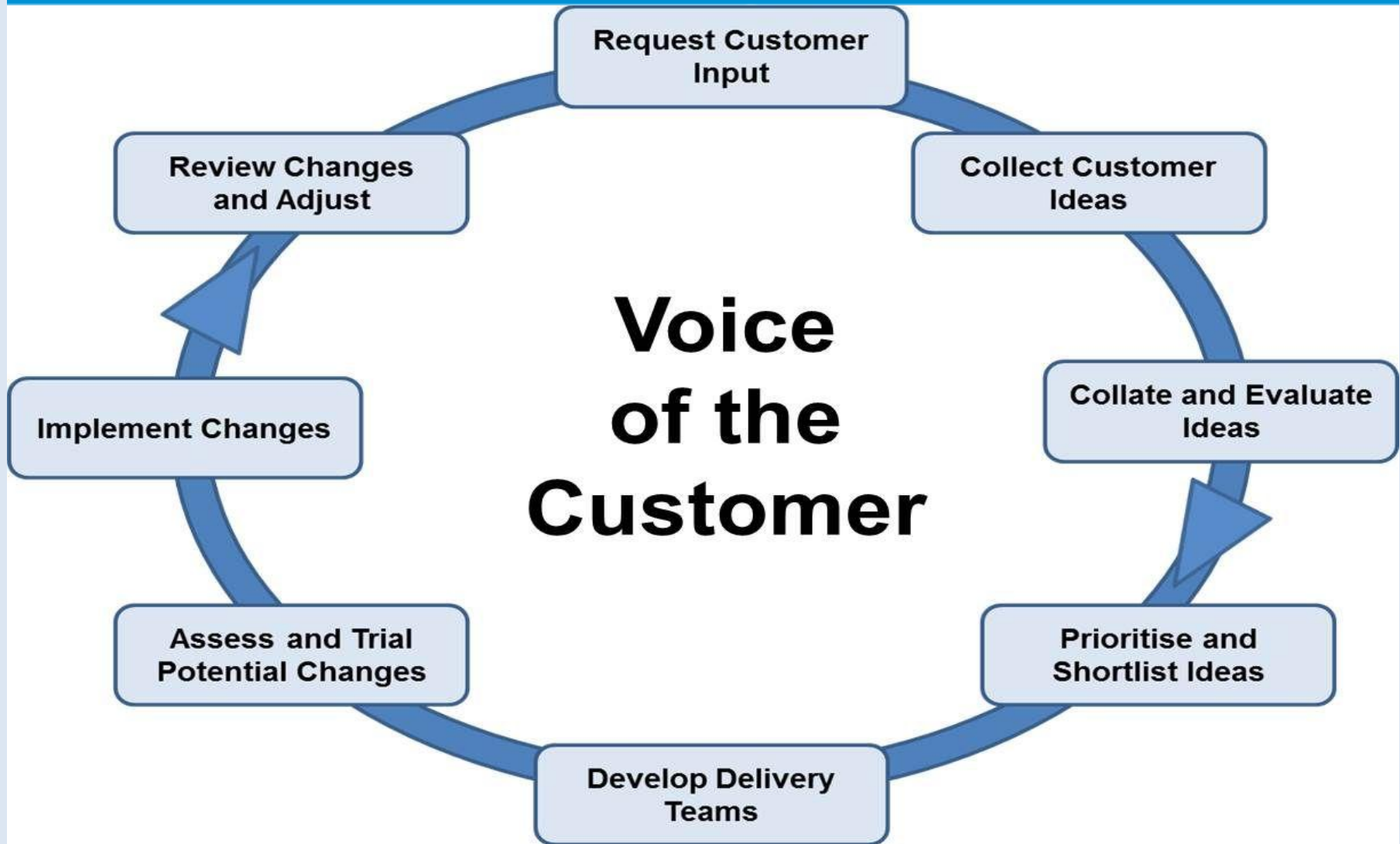


Swimlane Map

The Swimlane Map displays the process flow in separate lanes based on the group performing the step.



Voice of the Customer





Phase 2: Measure

Quantify the problem. How does the process currently perform? Or in other words, what is the magnitude of the problem? Measurement is critical throughout the life of the project. As the team starts collecting data they focus on both the process as well as measuring what customers care about. That means initially there are two focuses: reducing lead time or improving quality. In the Measure Phase, the team refines the measurement definitions and determines the current performance or the baseline of the process.

!! Steps

Determine How the Process
Currently Performs

Create a Plan to Collect the Data

Ensure the Data is Reliable

Gather the Baseline Data

Update Your Project Charter

🔧 Tools

Data Collection Plan

Operational Definitions

Check Sheet

Project Charter

Data Collection Plan

The Data Collection Plan is a roadmap detailing how to collect each piece of desired data.



Measure Name

What would you call it?



Data Type

Is it discrete or continuous?



Operational Definition

What is the airtight description?



Stratification Factors

Will you slice data by who, what, where and when?



Sampling Notes

How much data will you collect?

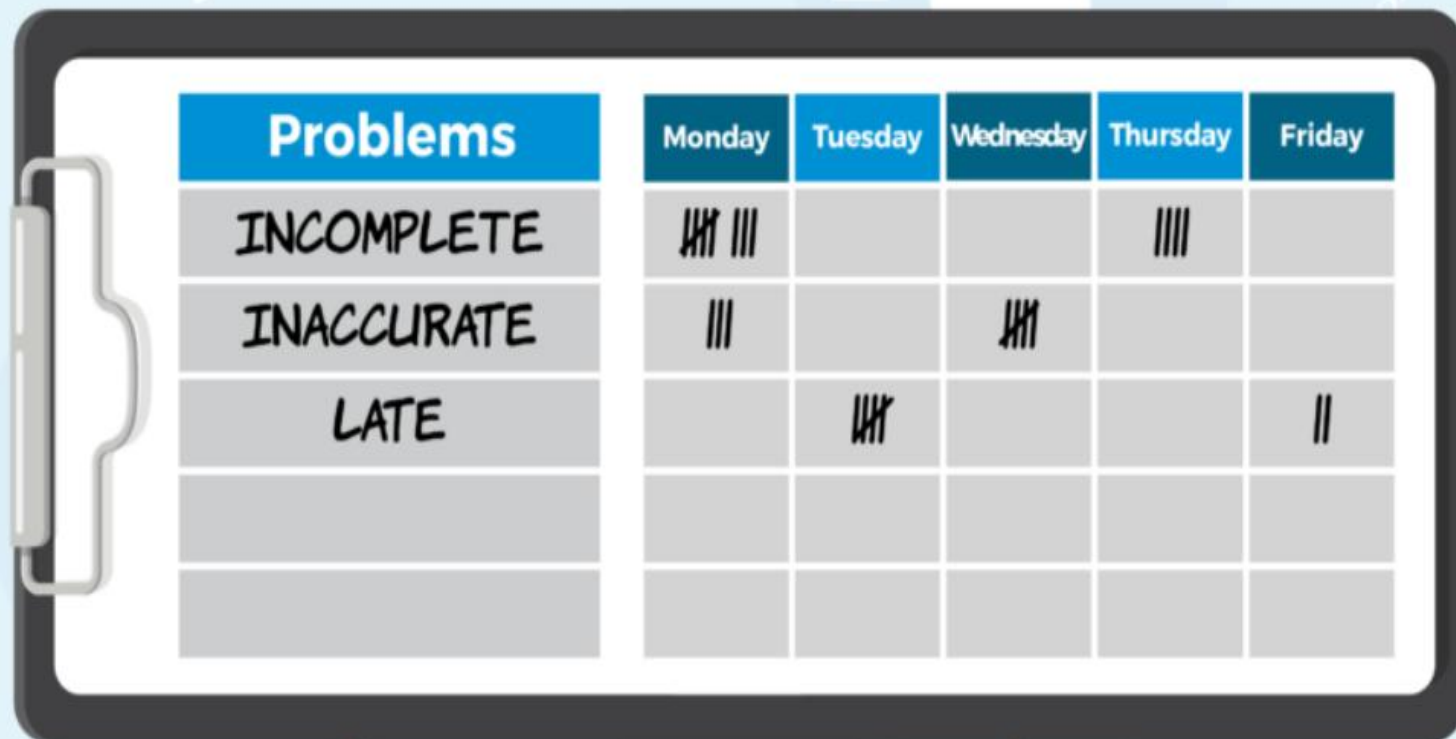


Who and How

Who is responsible and what method will they use?

Check Sheet

A Check Sheet is a customized tally sheet that enables manual collection of the nature and frequency of process issues.



Problems	Monday	Tuesday	Wednesday	Thursday	Friday
INCOMPLETE					
INACCURATE					
LATE					

Operational Definition

An Operational Definition is a detailed description that defines a measure to such a degree that everyone collects data the same way.

**OP
DEF**

of Sandwiches:

The total number of items with something delicious between two slices of bread

1, 2, 3... wait
a second!





Phase 3: Analyze

Identify the cause of the problem. What is causing the problem? The Analyze Phase is often not given enough attention and, without analysis, teams jump to solutions before knowing the true root causes of the issues. The result is teams who implement solutions but don't resolve the problem! These efforts waste time, consume resources, create more variation and, often cause new problems. The ideal is for teams to brainstorm potential root causes (not solutions), develop hypotheses as to why problems exist and then work to prove or disprove their hypotheses. Verification includes both process analysis and data analysis and has to be completed before implementing solutions. This is the crux of the Analyze Phase!

!! Steps

- Closely Examine the Process
- Graphically Display the Data
- Look for What Might be Causing the Problem
- Verify the Cause(s) of the Problem
- Update your Project Charter

🔧 Tools

Value Stream Map	Box Plots
Value Added Flow Analysis	Fishbone Diagram
Run Charts	5 Whys
Histograms	Root Cause Hypothesis
Pareto Charts	Project Charter

Value-Added Flow Analysis

The Value-Added Flow Analysis assesses which steps add value in the eyes of the customer and where time and effort are wasted in the process.



Value-Added Step

Time Spent on the Customer



Non-Value-Added Step

Wasted Time

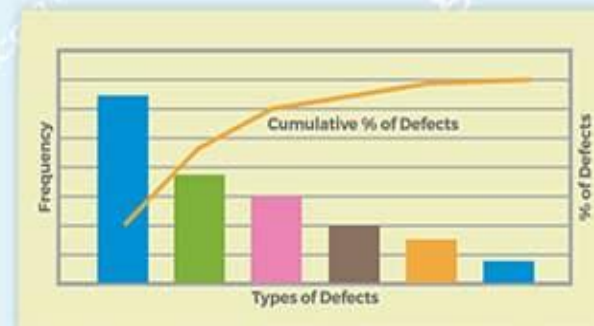
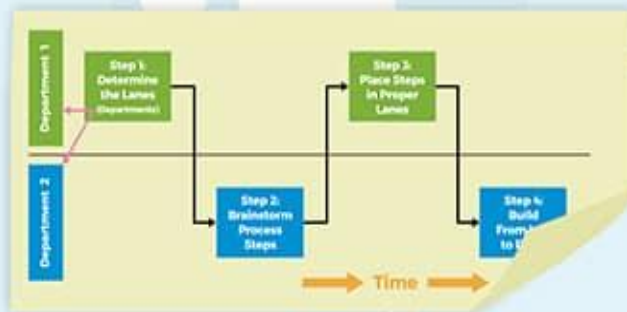
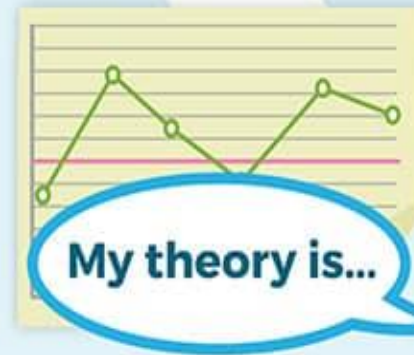
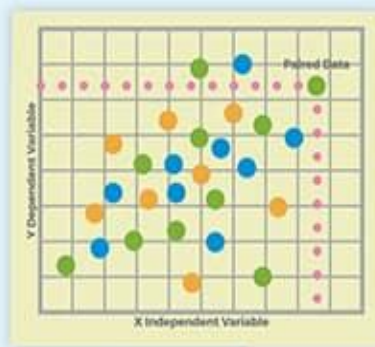


Non-Value-Added but Required Step

Unavoidable Wasted Time

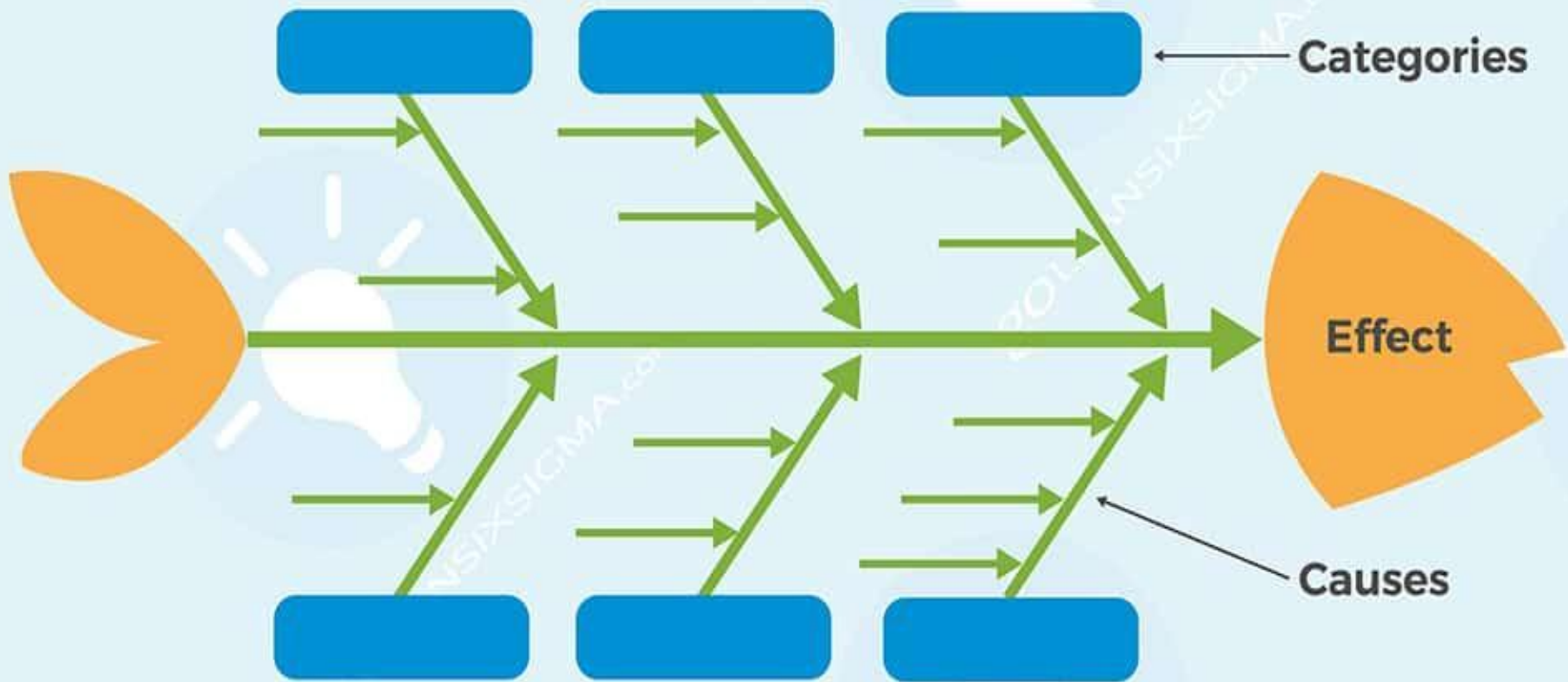
Root Cause Hypothesis Statement

A Hypothesis Statement is an educated guess about the suspected cause (or causes) of problems in a process.



Fishbone Diagram

A Fishbone Diagram is a structured brainstorming tool using categories to explore root causes for an undesirable effect.



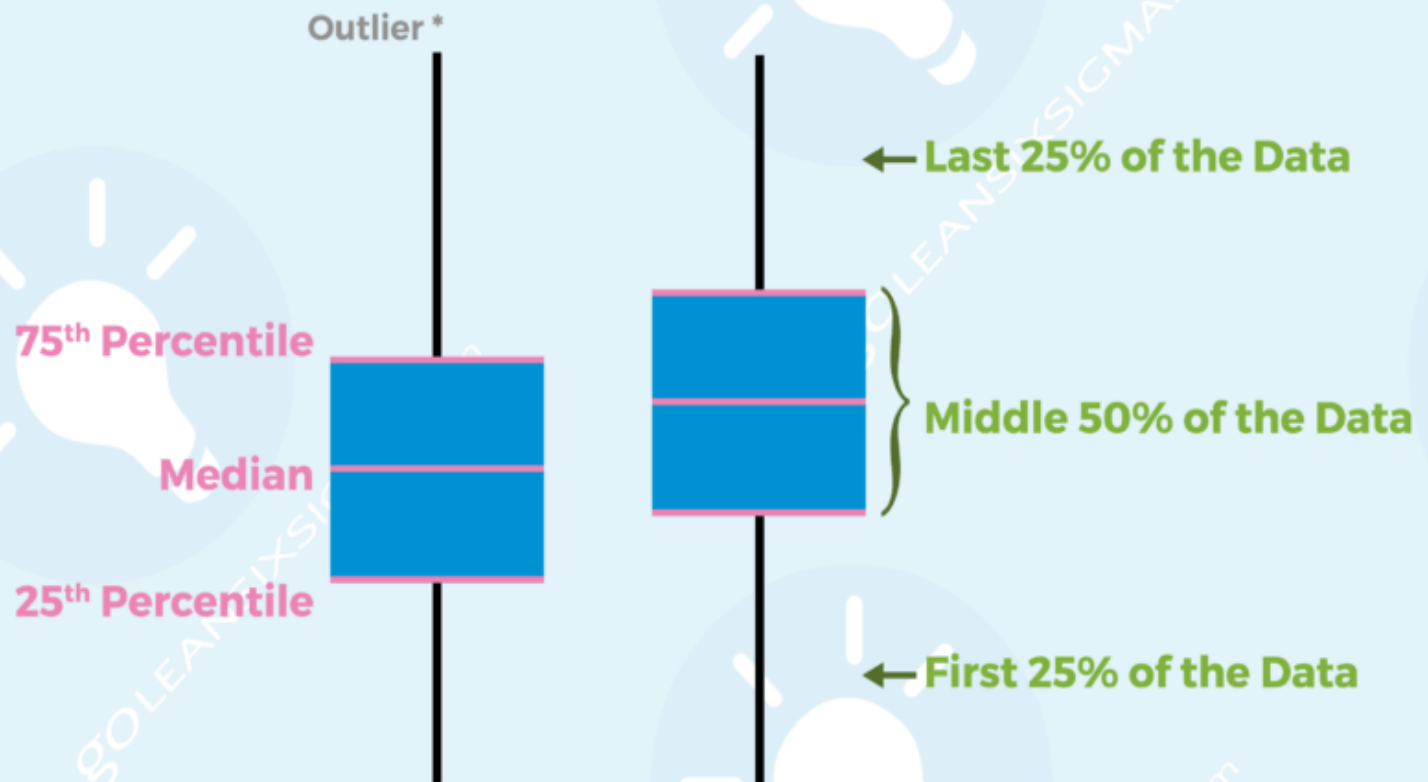
5 Whys

The 5 Whys is a simple analysis technique that moves past symptoms by asking “why” until reaching the true root cause of an issue.



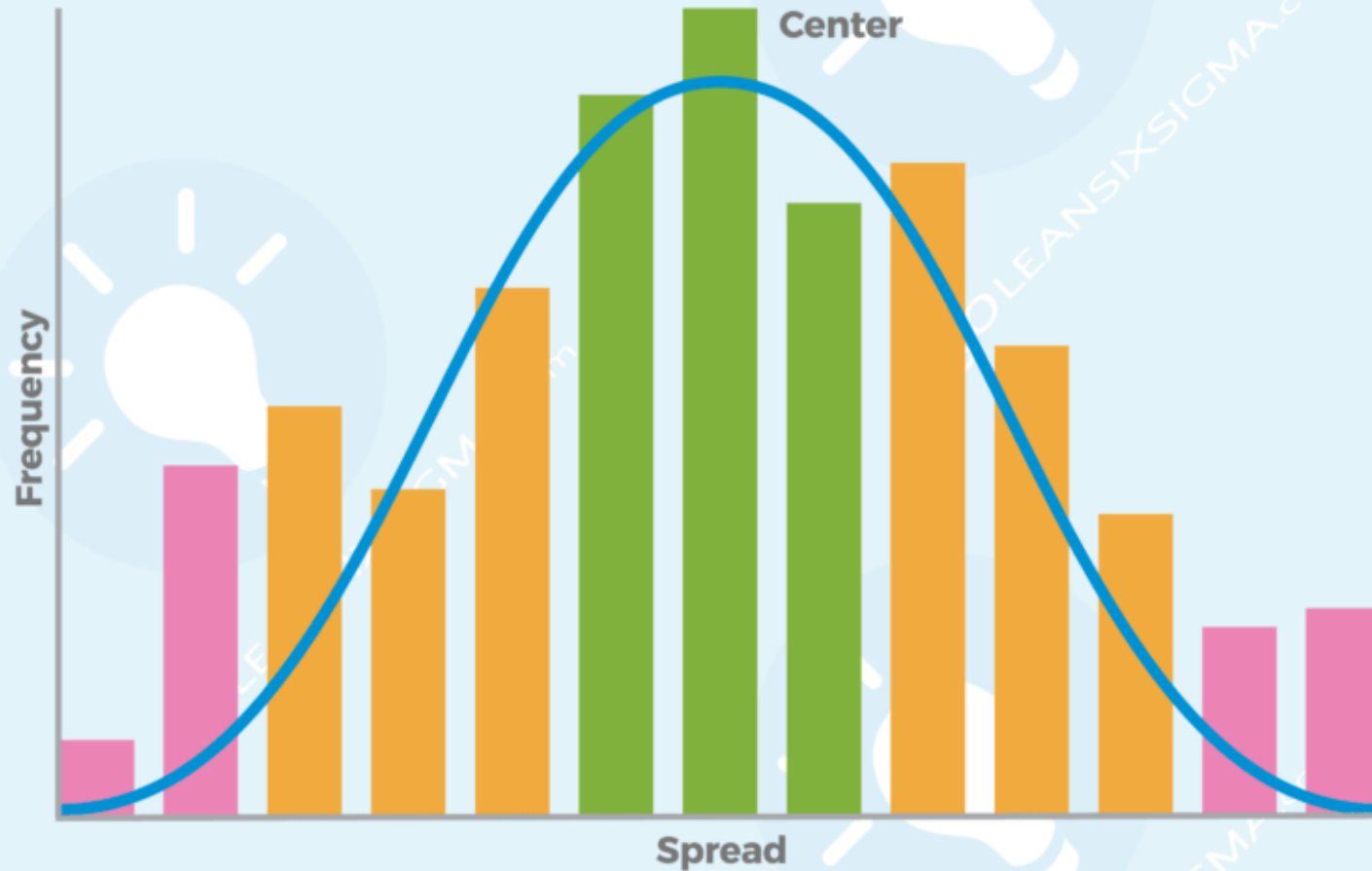
Box Plot

A Box Plot compares the distribution of different data sets by dividing them into fourths or “quartiles.”



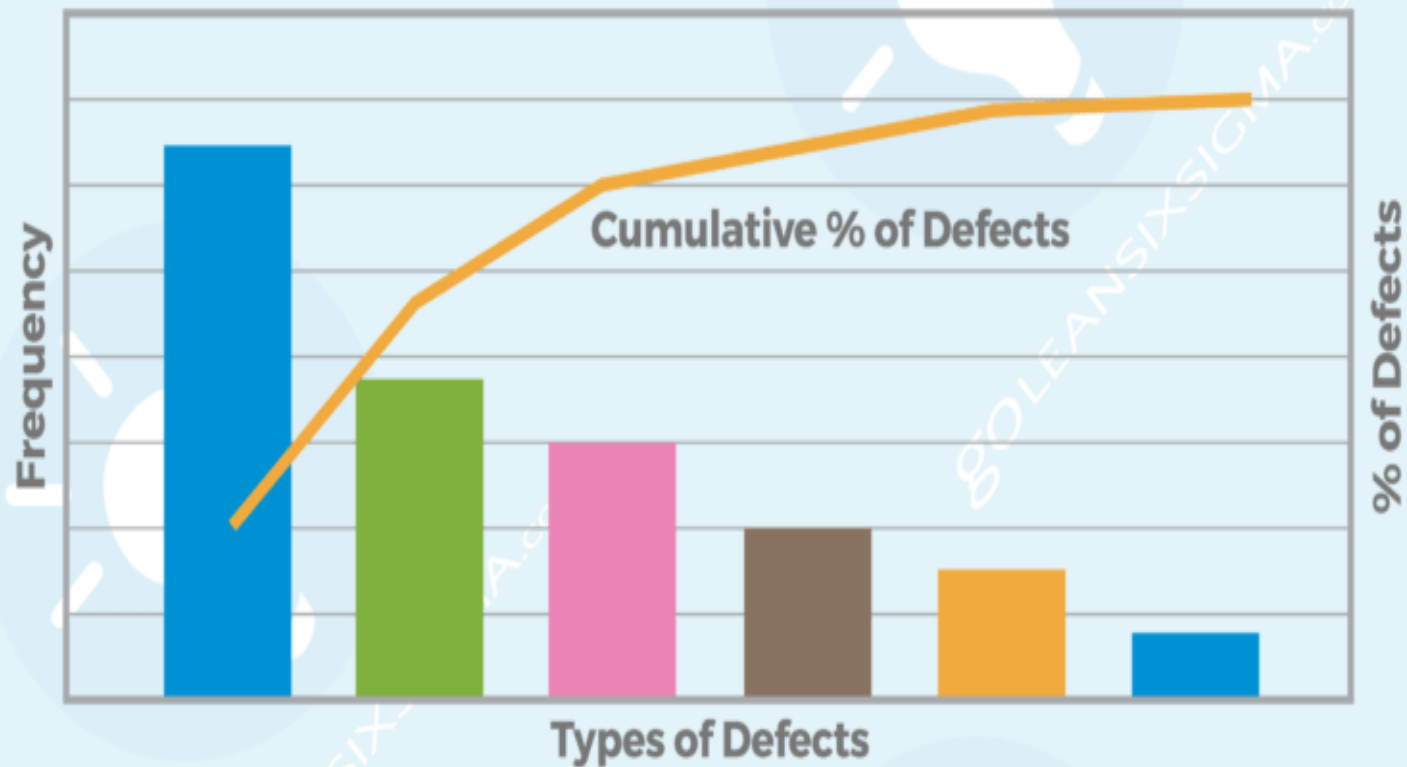
Histogram

Histograms show the distribution of values in a data set while highlighting the center, spread and shape of the data.



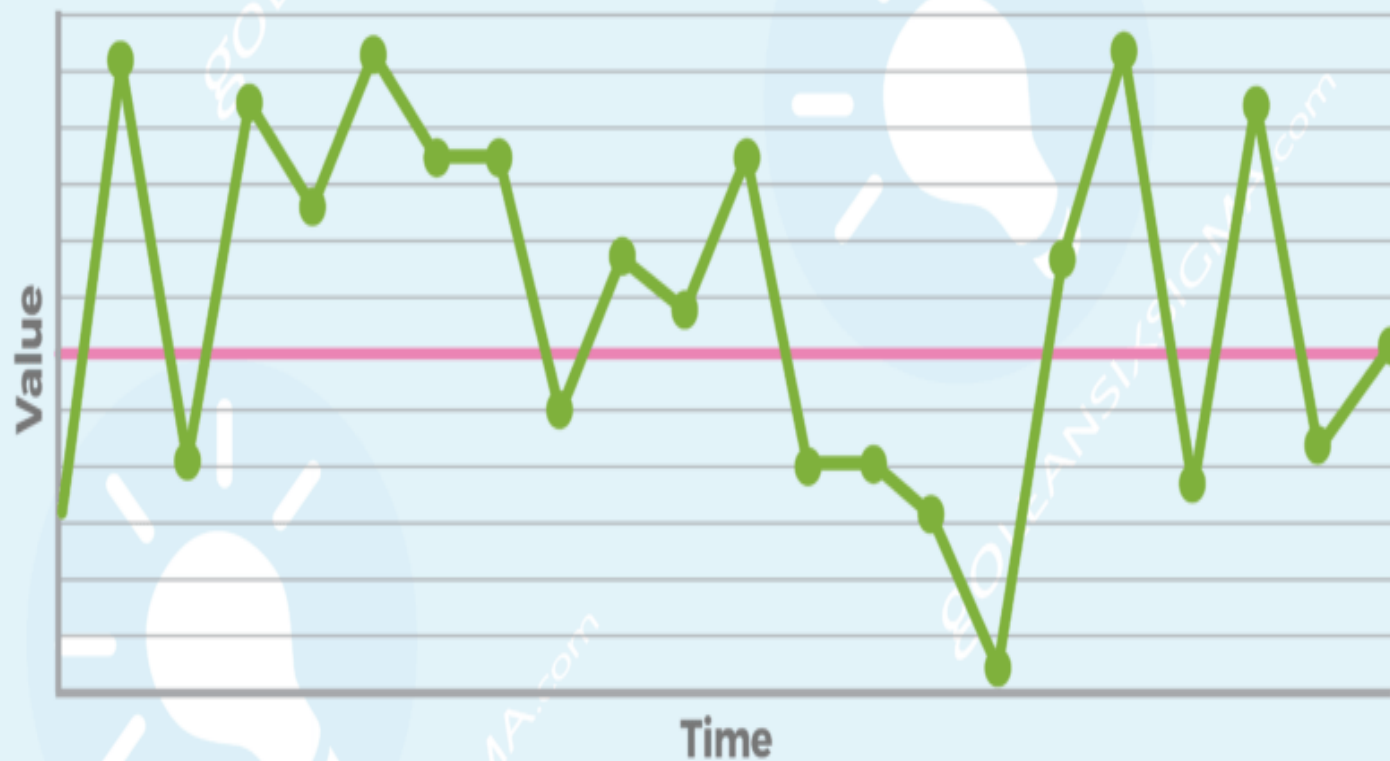
Pareto Chart

A Pareto Chart is a bar chart that displays the most significant types of defect occurrences in descending order.



Run Chart

A Run Chart is a time series plot that displays shifts and trends in data over time.





Phase 4: Improve

Implement and verify the solution. How will the team mitigate the root causes of the problem? Once the project teams have determined the root causes it's time to develop solutions. The Improve Phase is where the team brainstorms solutions, pilots process changes, implements solutions and lastly, collects data to confirm there is measurable improvement. A structured improvement effort can lead to innovative and elegant solutions that improve the baseline measure and, ultimately, the customer experience.

!! Steps

Brainstorm Solutions That Might Fix the Problem

Select the Practical Solutions

Develop Maps of Processes Based on Different Solutions

Select the Best Solution(s)

Implement the Solution(s)

Measure to Ensure Improvement

🔧 Tools

8 Wastes

5S

FSMA

Control Chart

Impact Effort Matrix

8 Wastes

The 8 Wastes are eight types of process obstacles that get in the way of providing value to the customer.



Defects

Efforts caused by rework, scrap, and incorrect information.



Overproduction

Production that is more than needed or before it is needed.



Waiting

Wasted time waiting for the next step in a process.



Non-Utilized Talent

Underutilizing people's talents, skills & knowledge.



Transportation

Unnecessary movements of products & materials.



Inventory

Excess products and materials not being processed.



Motion

Unnecessary movements by people (e.g., walking).



Extra-Processing

More work or higher quality than is required by the customer.

5S

5S is a five-step organization technique to create and maintain an intuitive workspace.



Sort

Keep only necessary items in the workplace.



Set In Order

Arrange items to promote efficient workflow.



Shine

Clean the work area so it is neat and tidy.



Standardize

Set standards for a consistently organized workplace.



Sustain

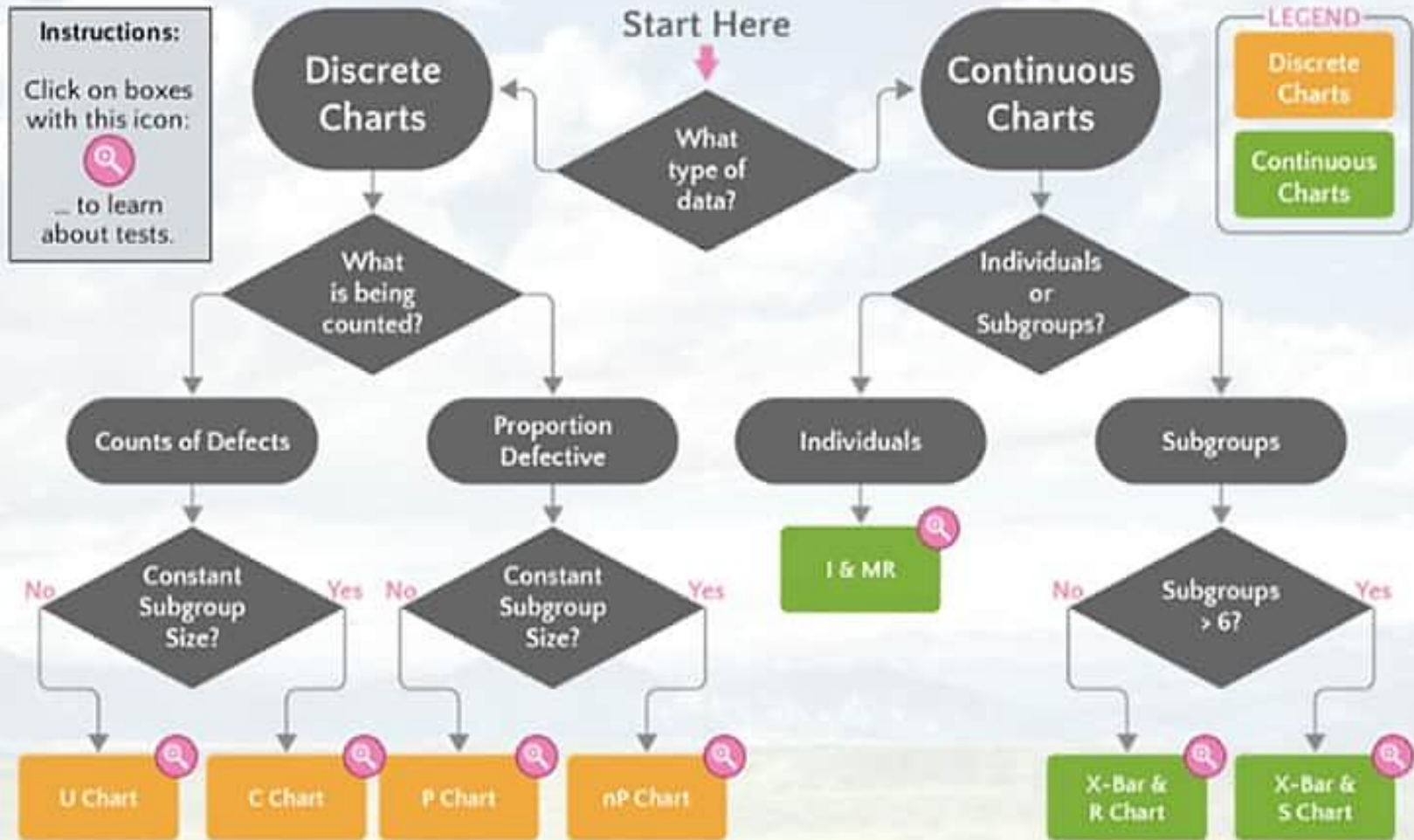
Maintain and review standards.

Failure Modes & Effects Analysis

Failure Modes & Effects Analysis (FMEA) is a risk management tool that identifies the influence of potential failures in a process in order to prevent them or create contingency plans.



Control Chart Test Decision Tree



Impact Effort Matrix Structure





Phase 5: Control

Maintain the solution. How do you sustain the improvement? Now that the process problem is fixed and improvements are in place, the team must ensure that the process maintains the gains. In the Control Phase the team is focused on creating a Monitoring Plan to continue measuring the success of the updated process and developing a Response Plan in case there is a dip in performance. Once in place, the team hands these plans off to the Process Owner for ongoing maintenance.

!! Steps

Ensure the Process is Properly Managed and Monitored

Document the Improved Process

Apply Improvements to Other Areas

Share and Celebrate Your Success

Continuously Improve the Process Using Lean Principles

🔧 Tools

Monitoring and Response Plan

Monitoring Plan Map

Executive Summary

Project Colure

Visual Mapping Checklist

Monitoring & Response Plan

The Monitoring & Response Plan checks whether a process is performing as expected and details what to do if not.



Executive Summary

The Executive Summary uses the A3 format to summarize the business case, the goal, the root causes, the improvements, and the benefits, and to display the “before” and “after” status of the Project Y.

Business Case

Root Cause Hypothesis

Solutions Implemented

Project Results

**Graphical Display
of Improvement**



Project Closure

Project Closure is the wrap-up of all the good that came of an improvement project and a reminder to continue. Continuous Improvement is the gift that keeps on giving!



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